

Competency Questions

- **List the rooms located on the first floor**

This information helps the building manager identify and categorize the spaces they manage on a specific level. Rooms on different floors might have different heating, cooling, or lighting requirements, and monitoring them based on location can be useful in assessing energy usage, occupancy, or maintenance needs.

- **Information about a room**

Detailed information about individual rooms is crucial for energy management and thermal comfort. The manager might need to know the room's dimensions, orientation

or purpose to optimize heating, cooling, or lighting systems. The manager might also need to identify the sensors and actuators present in the room to monitor and control environmental conditions effectively.

- **Walls adjacent to a room**

Understanding which walls are adjacent to a room is critical for energy efficiency analysis. Walls influence thermal behavior, especially in relation to insulation, material properties, or exposure to sunlight. This information can help in targeting areas for monitoring thermal leaks, which can reduce energy costs and improve comfort.

- **Details of a specific wall**

For thermal performance and maintenance purposes, it is important to know the material composition, the structure and the dimensions of specific walls.

- **Rooms adjacent to another room**

Identifying adjacent rooms can assist in understanding how environmental factors, such as temperature or humidity, propagate throughout the building. Heat transfer and air movement between neighboring rooms can affect overall energy efficiency and thermal comfort. This information is useful in configuring HVAC systems and

analyzing how certain rooms might influence the climate in others, enabling more effective thermal zoning and energy-saving strategies. It can also be beneficial for planning fire safety measures and evacuation routes.

- **Count the sensors in the building**

Knowing the sensors deployed in the building helps the manager keep track of the sensor network for operational monitoring and maintenance purposes. Sensors provide crucial real-time data on various parameters, such as temperature, humidity, and occupancy, and this information is essential for optimizing energy use and maintaining thermal comfort. The count ensures that all monitored spaces are covered, and that the building is equipped for effective energy and comfort management.

- **Count the humidity sensors in the building**

Humidity control is essential for occupant comfort and health, as well as for protecting the building from moisture-related damage, such as mold growth or material degradation. Knowing the number of humidity sensors helps the manager assess if the building is adequately monitored in terms of indoor air quality. This is particularly important in regions with variable climates or in spaces like restrooms or kitchens, where humidity levels fluctuate more significantly.

- **Where are located the humidity sensors**

The location of humidity sensors is critical for accurate monitoring and control. Sensors need to be placed in strategic locations, such as areas prone to moisture or spaces where air quality needs to be monitored closely. By knowing where humidity sensors are located, the building manager can ensure that coverage is sufficient and identify gaps that might need additional sensors for more comprehensive monitoring.

- **What is the temperature measured in a specific room**

Access to sensor measurements, such as room temperature, enables the building manager to directly monitor real-time operational data generated by the IoT infrastructure. This supports timely decision-making related to thermal comfort,

energy efficiency, and building performance.